# Dr. Babasaheb Ambedkar Technological University

## B. Tech. Electrical Engineering

### Summer Semester Examination-May-2018

**Subject: High Voltage Engineering [EE803]** **Mark: 70.**
**Semester: VIII** **Date: 15/05/2018.**

**Instructions: 1)** Question No.1 is compulsory. **2)** Solve the remaining Five questions. **3)** Necessary data given wherever required, if some data not given then it means it is part of Exam. **4)** Left side figure of the questions paper indicates the Marks of the questions.

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**Q. No. (1)** **Fill in the blanks with the appropriate answers.** [10]

**(a)** The best insulation among all insulating material available is-----------------------.

**(b)** Corona occurs before the breakdown in a sphere to ground air gap when ratio of gap distance to the radius of sphere is ---------------. (i) >1.0, (ii) >3.0, (iii) >10.

**(c)** Minimum sparking potential of air is about-----------. (i) 100 V, (ii) 4.4 kV, (iii) 325V.

**(d)** Which of the following gas is an electronegative gas...? (i) $O_2$, (ii) $SF_6$, (iii) Both(i) and (ii)

**(e)** The breakdown strength of mineral oil is about------------(i) 20 kV/mm, (ii) 50kV/mm,

**Q. No. (2)** **Solve any Two questions from the following** [12]

**(a)** Draw the experimental arrangement for the study of a Townsend discharge and explain different primary and secondary processes developing the ionization of gaseous medium.

**(b)** Explain the Townsends criterion for breakdown of gaseous medium.

**(c)** Discuss the various mechanism of vacuum breakdown.

**Q. No. (3)** **Attempt any Two questions from the following** [12]

**(a)** What are the different methods and means for purification of liquid dielectrics?

**(b)** A coaxial cylindrical capacitor is to be designed with an effective length of 20 cm. The capacitor is expected to have a capacitance of 1000 pF and to operate at 15 kV, 500 kHz. Select a suitable insulating material and give the dimensions of the electrodes.

**(c)** Explain the short-term and long-term breakdown of a composite dielectric material.

**Q. No. (4)** **Answer any Two questions from the following** [12]

**(a)** A Cockcroft-Walton type voltage multiplier has 8 stages with capacitances, all equal to 0.05 μF. The supply transformer secondary voltage is 125 kV at a frequency of 150 Hz. If the load current to be supplied is 5 mA, find (i) the percentage ripple, (ii) the regulation, and (iii) the optimum number of stages for minimum regulation or voltage drop.

**(b)** Why is a Cockcroft-Walton circuit preferred for voltage multiplier circuits? Explain its working with a schematic diagram along with waveforms.

**(c)** How are rectangular current pulses generated for testing purposes? How is their time duration controlled?

Q. No. (5) **Solve the following questions.** (Any Two) [12]

(a) A generating voltmeter has to be designed so that it can have a range from 20 to 200 kV dc if the indicating meter reads a minimum current of 2 μA and maximum current of 25 μA, what should the capacitance of the generating voltmeter be?

(b) What is capacitance voltage transformer? Explain with phasor diagram how a tuned capacitance voltage transformer can be used for voltage measurement in power systems.

(c) A 3-phase single circuit transmission line is 400 km long. if the line is rated for 220 kV and has the parameters, R=0.1Ω/km, L= 1.26 mH/km, C=0.009μF/km, and G=0, find (i) the surge impedance, and (ii) the velocity of propagation neglecting the resistance of the line. If a surge of 150kV and infinitely long tail strikes at one end of the line, what is the time taken for the surge to travel to the other end of the line?

Q. No. (6) **Attempt the following questions.**(Any Two) [12]

(a) A transmission line of 500Ω surge impedance is connected to a cable of 60Ω surge impedance at the other end. If a surge of 500kV travels along the line to the junction point. Find the voltage build-up at the junction?

(b) What is a surge arrester? Explain its function as a shunt protective device.

(c) Write a short note on testing of a bushing as per the IS standard.